LESSON PLAN

NBCD (30)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration of real time data acquisition system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration of real time data acquisition system.
5. Specific Objectives	30.1.1.1 Trainee should be able to practical on various lab FTL, RF & Embedded lab.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Hands-on demonstration of DAQ system interfacing with sensors, safety checks on high voltage lines, and instructor supervision during signal injection.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a data acquisition system?
(b) Name three types of sensors used in FTL labs.
(c) Why is real-time data acquisition critical in RF environments?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to DAQ hardware components • Signal conditioning and sensor interfacing • Real-time data processing algorithms • Safety protocols for embedded labs	L & Ds	PPT, IB	15 min
(b)	• Configuration of FTL lab DAQ modules • Calibration of RF signal inputs • Troubleshooting common acquisition errors • Data logging and visualization techniques	L & Ds	PPT, PDF	20 min
(c)	• Integration of embedded lab sensors • Analysis of acquired waveform data • Reporting standards for lab results • Maintenance of DAQ equipment	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of signal conditioning in DAQ systems.
(b) How do you verify the accuracy of real-time data acquisition?
(c) What steps are taken to ensure safety during embedded lab operations?

12. Home Assignment and Follow Up	Prepare a brief report on the differences between FTL and RF lab DAQ configurations for the next lesson.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Operation of oscilloscope and function generator and its controls and observe waveform, measurement of various parameters of waveform.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min